DGCA CPL / ATPL STUDY NOTES • RADIO NAVIGATION

✈ Chapter 9: ILS

Instrument Landing System

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1. Introduction

The Instrument Landing System (ILS) is the most accurate approach and landing aid in current use. It has been in existence for over 40 years and is the ICAO standard precision approach system.

📡 Key Characteristics

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- **Precision approach**

  system — provides guidance in

  both

  horizontal and vertical planes
- Enables automatic landings (autoland) in Cat III conditions
- Guides the pilot from the coverage limit down to

  Decision Height (DH)
- At DH: land (if required visual references exist) or execute missed approach

ILS provides the pilot with visual instructions in the cockpit to enable him to fly the aircraft down a predetermined glide path and extended runway centre line (localizer) to his Decision Height (DH). At decision height the pilot decides to land (if he has the required visual references and sufficient room to manoeuvre the aircraft for a safe touchdown) or he goes around (overshoots) and carries out the published missed approach procedure.

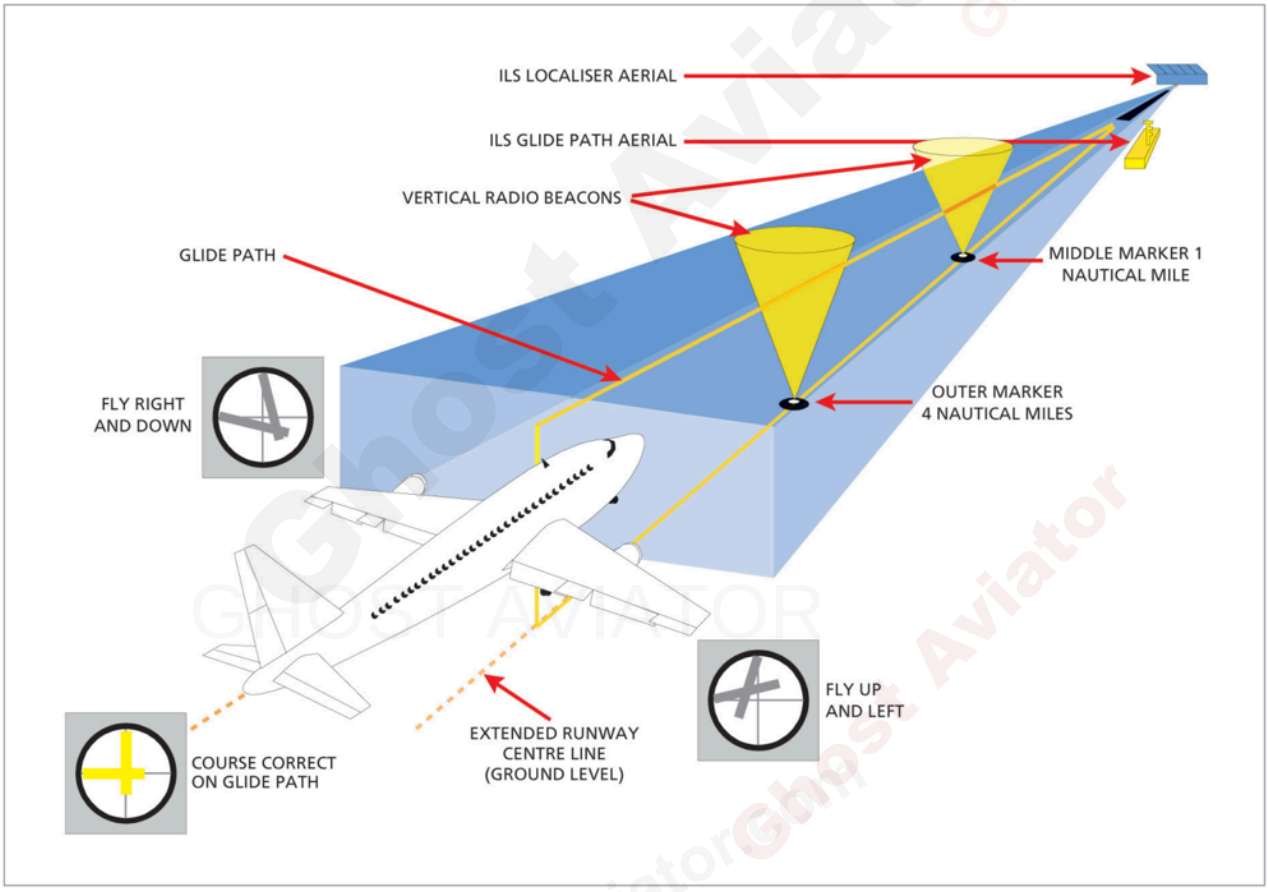


Figure 9.1 The Instrument Landing System (ILS)

Fig 9.1: The Instrument Landing System (ILS) — overview of all components

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2. ILS Components

📡 Ground Installation: Three Main Components

Component	Function	Location	Band
Localizer (LLZ)	Lateral (left/right) guidance — runway centreline	~300 m from upwind end of runway	VHF
Glide Path (GP)	Vertical (up/down) guidance — glide slope	300 m from threshold, ~200 m from runway edge abeam touchdown	UHF
Marker Beacons	Range check — cross-check height vs distance	OM, MM, IM at set distances from threshold	VHF 75 MHz

✓ Optional/Associated Components

- **Locator (L)**

  : Low-powered NDB co-located at the OM beacon site
- **DME**

  : Frequency-paired with ILS; zero-referenced to runway threshold; supplements or replaces marker range information
- **Back Course**

  : Non-precision approach on back beam of localizer (approved in some countries; **NOT permitted in UK**

)

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3. ILS Frequencies & Pairing

LOCALIZER (VHF)

108 – 111.975 MHz

40 channels, odd tenths

GLIDE PATH (UHF)

329.15 – 335 MHz

40 channels, freq. paired

MARKERS (ALL)

75 MHz

OM + MM + IM

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⚡ Localizer Frequency Sharing

In 108–112 MHz:

Odd tenths

= ILS Localiser (108.1, 108.3, 108.5…);

Even tenths

= VOR. Each ILS frequency also has a second step: e.g., 108.1 AND 108.15 for ILS.

📡 Frequency Pairing Advantages

- Selecting the localizer frequency **automatically activates the GP receiver**
- Reduces pilot workload — one selection activates both
- Reduces potential for wrong frequency selection
- Only one identifier needed for both localizer and glide path

Example: LLZ 108.1 MHz is paired with GP 334.7 MHz; LLZ 111.95 MHz paired with GP 330.95 MHz.

DME Paired with ILS

📡 ILS/DME

- DME ranges are zero-referenced to the **ILS runway threshold**
- Protected within ILS localizer service area up to **25,000 ft**
- May also be used for published SIDs and STARs (with extended coverage when notified)
- Do not use DME outside stated limits — may give errors

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4. Identification & Monitoring

📻 ILS Identification

- 2 or 3-letter Morse code;

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first letter is usually "I"

- Transmitted at **7 groups per minute**
- Only one ident needed (shared by localizer and GP)
- Ident automatically suppressed if ILS becomes **unserviceable**
- During maintenance: ident replaced by **continuous tone** or removed completely

⚠ Critical Rules

- If ident is a **continuous tone** → ILS under maintenance → **DO NOT USE**
- If ident **ceases** during approach → ILS is unserviceable → **discontinue immediately**
- Unserviceable glide path only: ATC will warn; **do not use GP**

Ground Monitoring

🔍 Automatic Monitor Triggers Action When:

Parameter	Cat I Limit	Cat II Limit	Cat III Limit
Localizer shift from centreline	> **35 ft**	> **25 ft**	> **20 ft**
Glide path angle change	> **0.075 × basic GP angle**		
Power reduction	> **50%**		

Actions: Cease radiation / Remove ident & nav components / Lower category (Cat II/III may revert to Cat I/II).

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5. Marker Beacons

Markers transmit at **75 MHz** in a narrow **fan-shaped vertical beam**. The signal is received only when the aircraft flies through the fan (not directional). Reception is indicated by synchronous aural tones and cockpit lights.

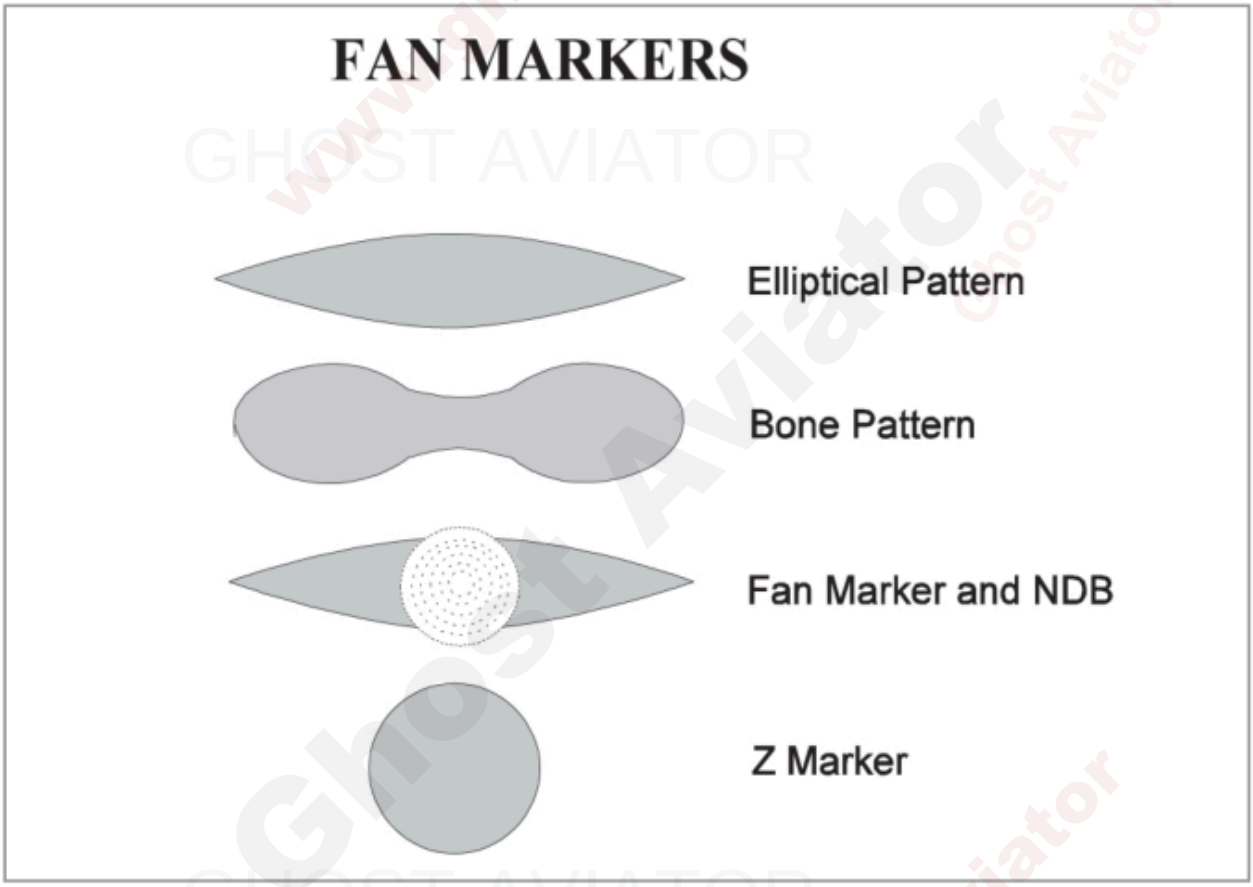


Figure 9.2 Marker beacon radiation patterns

Fig 9.2: Marker beacon radiation patterns — fan-shaped vertical beams

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The radiation patterns for ILS marker beacons is vertical and appears lens shaped or bone shaped in plan view. *Figures 9.2* and *9.3* show the horizontal and vertical profiles of ILS marker beacons. The signal is only received if the aircraft is flying within the fan; it is not a directional aid. Reception is indicated by synchronous aural identifiers and lights as shown in the following table.

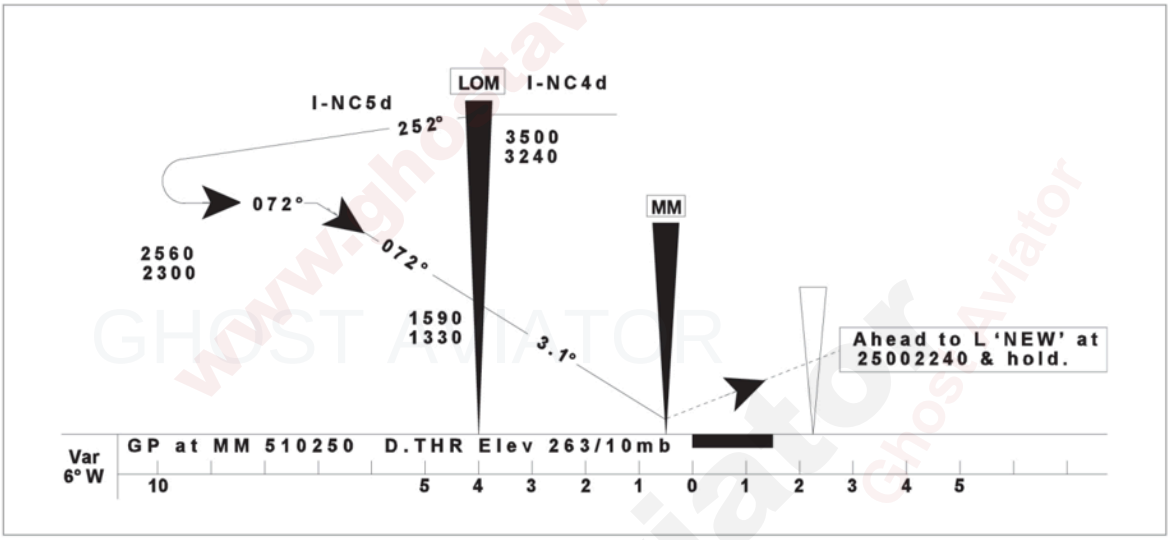


Figure 9.3 The outer marker and the middle marker

	Cockpit Light	Ident	Modulating Frequency	Pitch	Touchdown Range
OM	**BLUE**	2 dashes/sec	400 Hz	**Low**	6.5 – 11.1 km (3.5 – 6 NM)
MM	**AMBER**	Alternate dots and dashes 3/sec	1300 Hz	**Medium**	1050 m ± 150 m (3500' ± 500')
IM	**WHITE**	6 dots/sec	3000 Hz	**High**	75 - 450 m (250′ – 1500′)

Fig 9.3: OM and MM positions, plus marker identification table

Marker Beacon Summary

Marker	Cockpit Light	Ident Signal	Audio Freq	Range from Threshold

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OM (Outer)	**BLUE**	2 dashes/sec	**400 Hz** (Low pitch)	6.5 – 11.1 km (3.5 – 6 NM)
MM (Middle)	**AMBER**	Alt dots & dashes 3/sec	**1300 Hz** (Medium pitch)	1050 m ± 150 m (3500′ ± 500′)
IM (Inner)	**WHITE**	6 dots/sec	**3000 Hz** (High pitch)	75 – 450 m (250′ – 1500′)

⚡ Memory Aid — Markers

B

ig

B

lue

B

eacon =

O

uter Marker (3 Bs, low dashes) —

A

mber = Middle (Alternating) —

W

hite & high = Inner (Whitest, Highest pitch, fastest dots)

✓ Z Markers

Z markers have a

cylindrical

vertical radiation pattern. Used to mark airway reporting points or co-located with NDBs. They indicate when the aircraft is overhead (compensating for the NDB cone of silence).

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6. ILS Coverage

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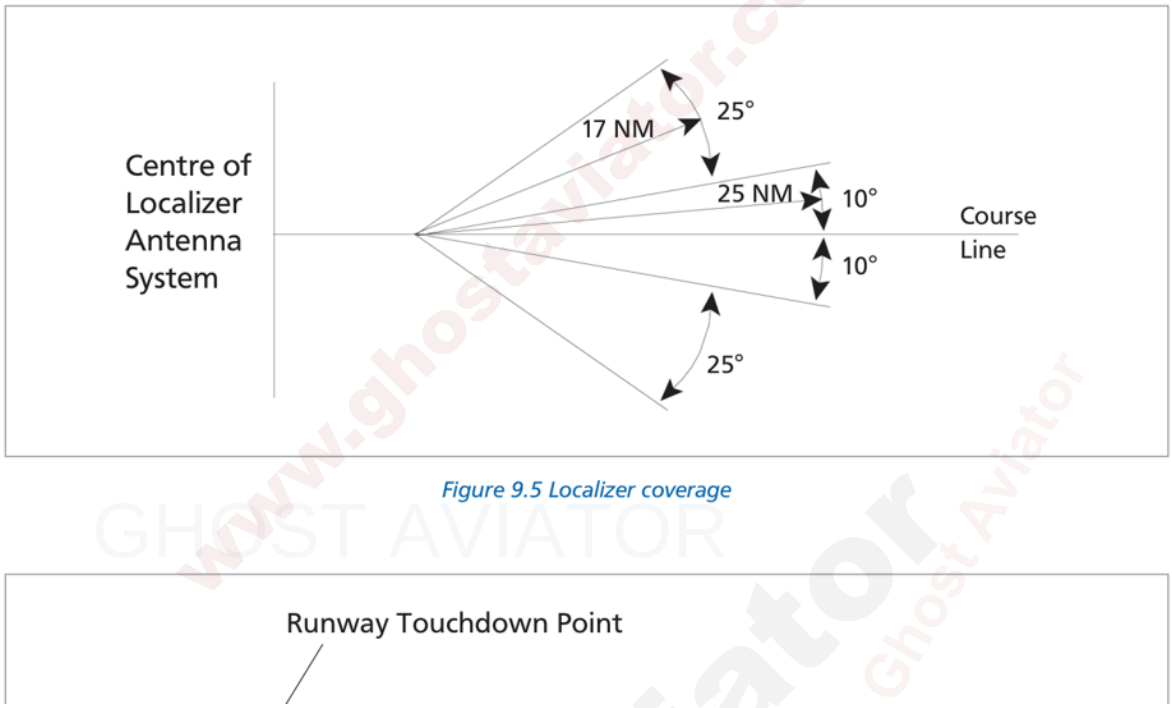


Figure 9.5 Localizer coverage

Fig 9.5: Localizer coverage sectors — 25 NM within ±10°, 17 NM within ±35°

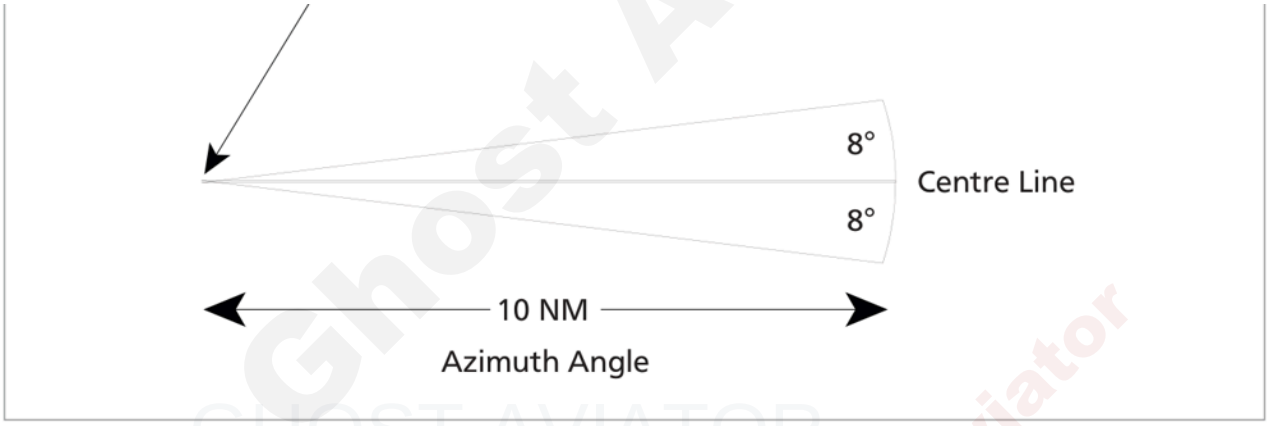


Figure 9.6 Glide path horizontal coverage

Fig 9.6: Glide path horizontal coverage — 10 NM within ±8° azimuth

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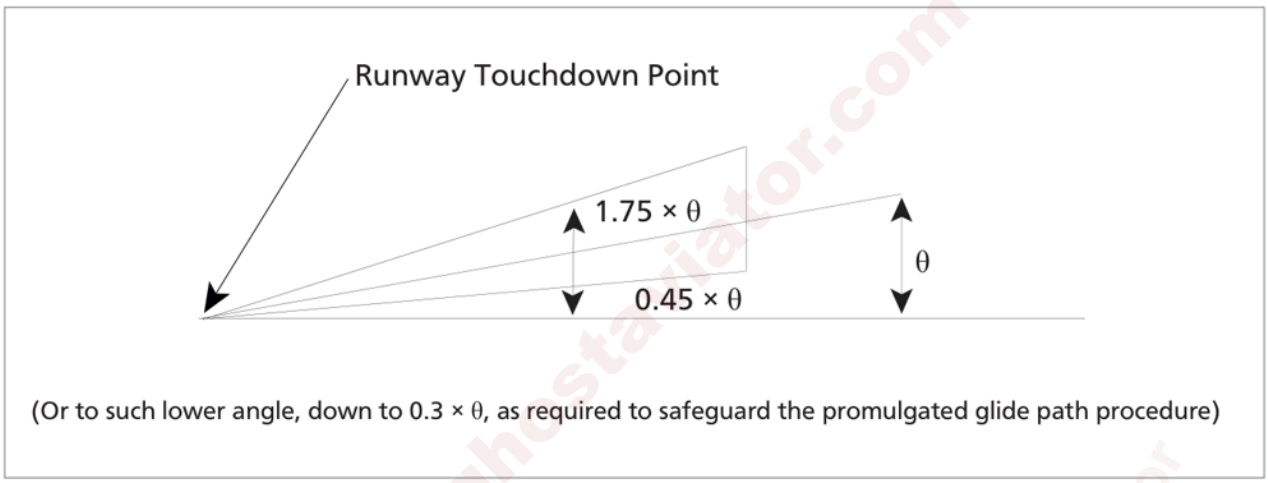


Figure 9.7 Glide path vertical coverage

Fig 9.7: Glide path vertical coverage — 0.45θ to 1.75θ above horizontal

📐 Localizer Coverage

Sector	Range
Within ±10° of centreline	**25 NM** (46.3 km)
10° to 35° from centreline	**17 NM** (31.5 km)
Outside ±35°	**10 NM** (18.5 km) if provided

Note: May be reduced to 18 NM (±10°) when alternative navaids provide satisfactory intermediate approach coverage.

📐 Glide Path Coverage

Parameter	Value
Azimuth coverage	**±8°** either side of centreline
Range	At least **10 NM**
Vertical — upper limit	**1.75 × θ** above horizontal
Vertical — lower limit	**0.45 × θ** (may reduce to 0.3θ if required)

Where θ = promulgated glide path angle (usually 3°).

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⚡ Outside Coverage Sectors

If an aircraft uses the ILS outside published coverage sectors, it may receive **false on-course or reverse sense signals**. The ILS cannot guarantee correct indications outside its coverage area.

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7. Principle of Operation

7.1 Localizer

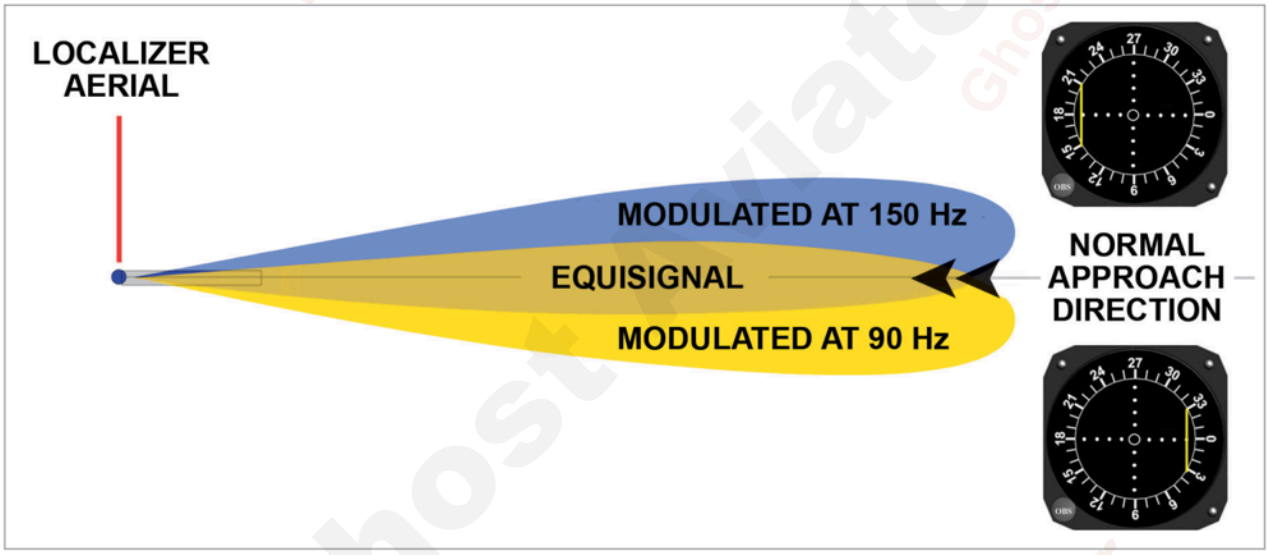


Figure 9.8 *Localizer radiation pattern*

Fig 9.8: Localizer radiation pattern — two overlapping lobes with 90 Hz and 150 Hz modulation

📡 Localizer Operation

Sector	Frequency	Colour Code	CDI Indication
Left-hand lobe (aircraft left of %)	**90 Hz**	Yellow	Needle deflects RIGHT → Fly Right
Right-hand lobe (aircraft right of %)	**150 Hz**	Blue	Needle deflects LEFT → Fly Left

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On centreline	DDM = 0	—	Needle centred

DDM (Difference in Depth of Modulation) increases with distance from the centreline. The DDM drives the vertical needle of the CDI/HSI.

⚡ **Memory Aid — Localizer Lobes**

"

150 = Blue = Right = Fly Left

" (Right of centre → fly left to return to centre)

"

90 = Yellow = Left = Fly Right

"

7.2 Back Course ILS

⚠ **Back Course Disadvantages**

- Glide path indications are **incorrect** (would guide to wrong end of runway)
- CDI needle sense is **REVERSED** for back course approach (HSI gives correct sense if front QDM selected)
- **No range-check markers** available
- Not permitted in the UK — ignore any back course indications in UK

7.3 Glide Path

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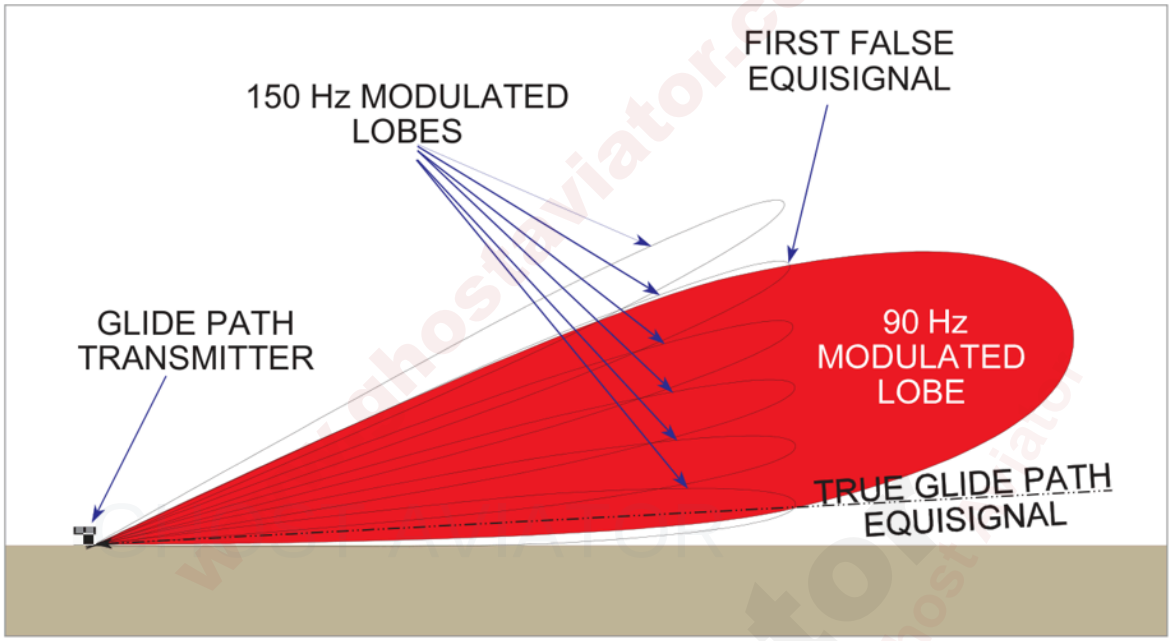


Figure 9.9 Glide path radiation pattern

Glide Slope

The glide slope UHF transmitter is located to one side of the runway approximately 200 m from the runway edge, 300 m upwind of the threshold.

The same principle is used as for the localizer, but a UHF carrier wave is used and the lobes

Fig 9.9: Glide path radiation pattern — two lobes in vertical plane, with false glide slope above

Glide Path Operation

Lobe	Modulation	Aircraft Position	CDI Indication
Upper lobe (large)	**90 Hz**	Aircraft above GP	GP needle deflects DOWN → Fly Down
Lower lobe (small)	**150 Hz**	Aircraft below GP	GP needle deflects UP → Fly Up
On glide path	DDM = 0	On the GP	GP needle centred

Standard GP angle: **3°** (ICAO requirement: 2° – 4°).

GP transmitter: UHF, located ~300 m upwind of threshold, ~200 m from runway edge.

7.4 False Glide Slopes

⚠ False Glide Slopes

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- Caused by metallic structure reflections and antenna height/propagation characteristics
- First false GP at approximately
  twice the promulgated glide path angle
- For standard 3° GP: first false GP at approximately **6°**
- False GPs always occur
  above
  the true glide slope
- Normal practice: intercept glide slope
  from below
- At Heathrow: continuous descent approach may capture GP from above — confirm height with DME/markers

✓ ILS Reference Datum Point

Located vertically above the intersection of the runway centreline and threshold at a specified height (~50 ft). The downward extended portion of the ILS glide path passes through this point. Height is published in the UK AIP, AD section.

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8. Displays & Interpretation

8.1 CDI (Course Deviation Indicator) in ILS Mode

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ILS Reference Datum Point

The ILS reference datum point is a point at a specified height (around 50 feet) located vertically above the intersection of the runway centre line and threshold, through which the downward extended portion of the ILS glide path extends. This value is to be found in the remarks column for the particular airfield in the UK AIP, AD section.

Visual Glide Path Indicators

The approach light systems such as PAPIs give a visual indication of the glide path to the runway that would be the same as that for the ILS so that during the final phase of the approach the pilot should get similar indications of glide path from both systems. However the visual indications are designed for a mean eye height (meht) of the pilot and they would therefore vary slightly since the pilot's position will vary depending upon the size of the aircraft.

ILS Presentation and Interpretation

Indicators

Localizer and glide path information can be displayed:

- on a Course Deviation Indicator (CDI) or
- on the Horizontal Situation Indicator (HSI).

Interpretation of a CDI display is shown in *Figure 9.10*. The HSI display is shown at *Figure 9.11*. The main difference to note is that on the HSI there is a course selector which should be set on the QDM of the runway. The deviation indications then appear in the correct sense.

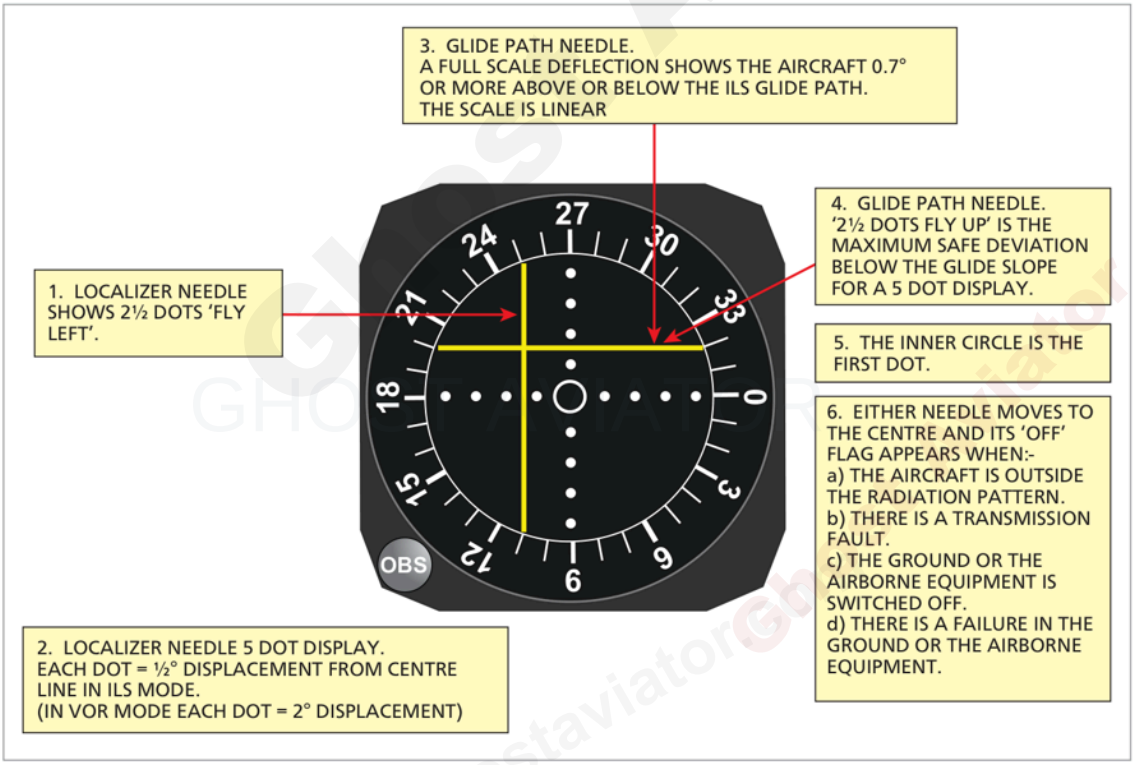


Figure 9.10 ILS course deviation indicator

Fig 9.10: ILS CDI — localizer needle (vertical) and glide path needle (horizontal) with annotations

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📐 CDI Sensitivity in ILS Mode

Parameter	Localizer	Glide Path
Sensitivity	**0.5° per dot**	**0.14° per dot**
Full Scale Deflection	**±2.5°** (= 5 dots)	**±0.7°** (= 5 dots)
Max safe deviation below GP	—	**2.5 dots fly up** (= 0.35°)

Compare with VOR mode:

CDI sensitivity = 2°/dot in VOR mode, 0.5°/dot in ILS mode. 5× more sensitive!

⚠ Maximum Safe Deviations

- If left/right localizer deviation exceeds
  half full scale (>2.5 dots)
  → initiate
  immediate go-around
- If glide path fly-up exceeds
  2.5 dots fly up
  → initiate
  immediate go-around
  (terrain clearance may be compromised)

8.2 HSI (Horizontal Situation Indicator) in ILS Mode

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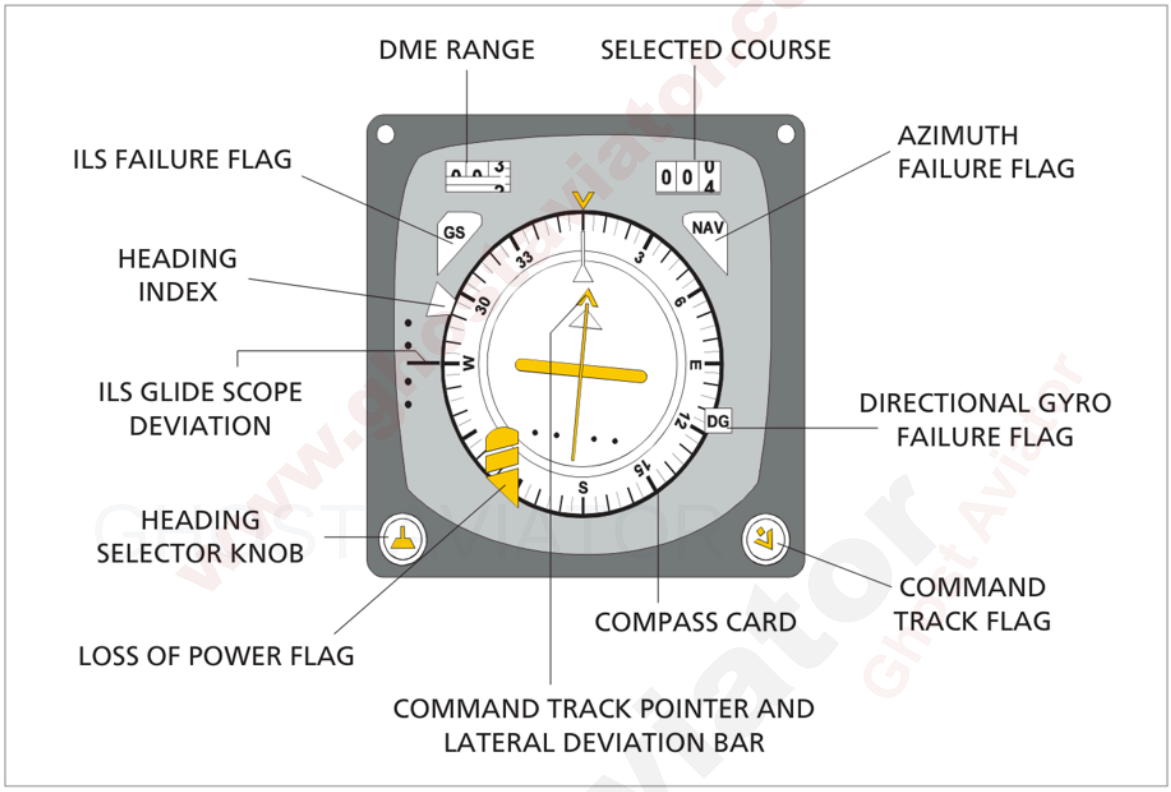


Figure 9.11 A typical HSI

Localizer Indications

Front course approach indications for fly left and right are shown in *Figure 9.12*. Full scale deflection of the needle indicates that the aircraft is 2.5° or more left or right of the centre-line i.e. the sensitivity is 0.5° per dot.

Fig 9.11: A typical HSI showing ILS components — lateral deviation bar and GP needle

HSI in ILS Mode

- Course selector should be set to the **QDM of the runway**
- Deviation bar and GP needle give **correct sense indications**
- HSI gives correct sense even on back course (if front course QDM is set)
- Azimuth failure flag, ILS failure flag, and loss of power flag visible

8.3 Localizer On-Course Indications

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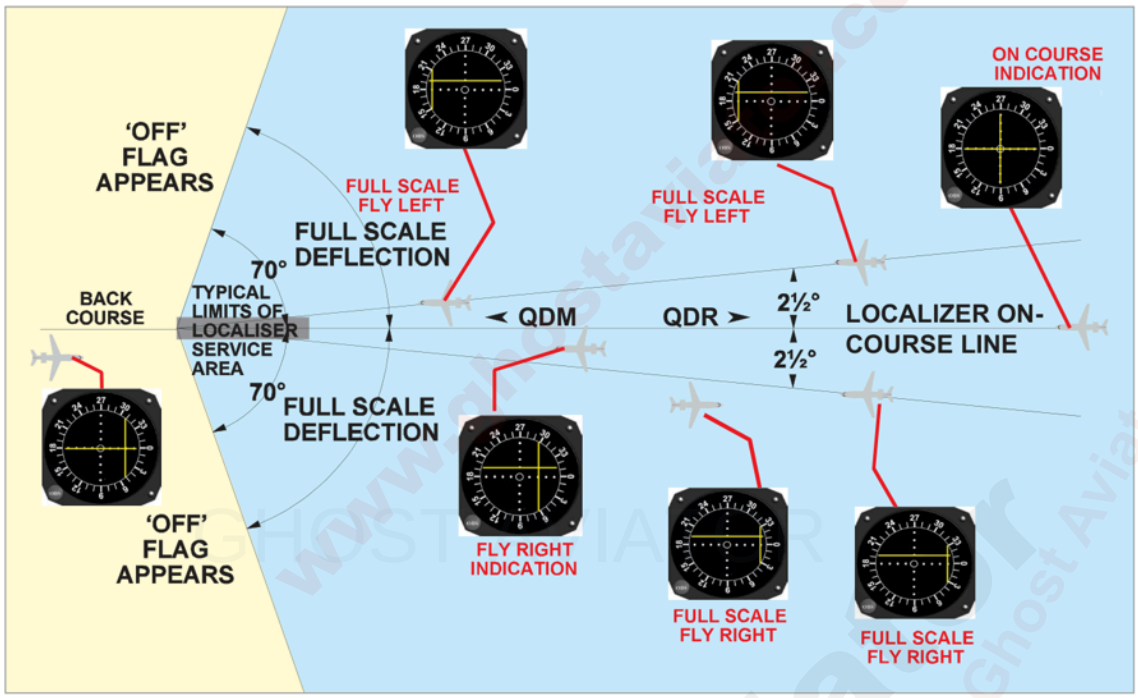


Figure 9.12: Localizer

Fig 9.12: Localizer on-course line indications — fly left/fly right CDI deflections

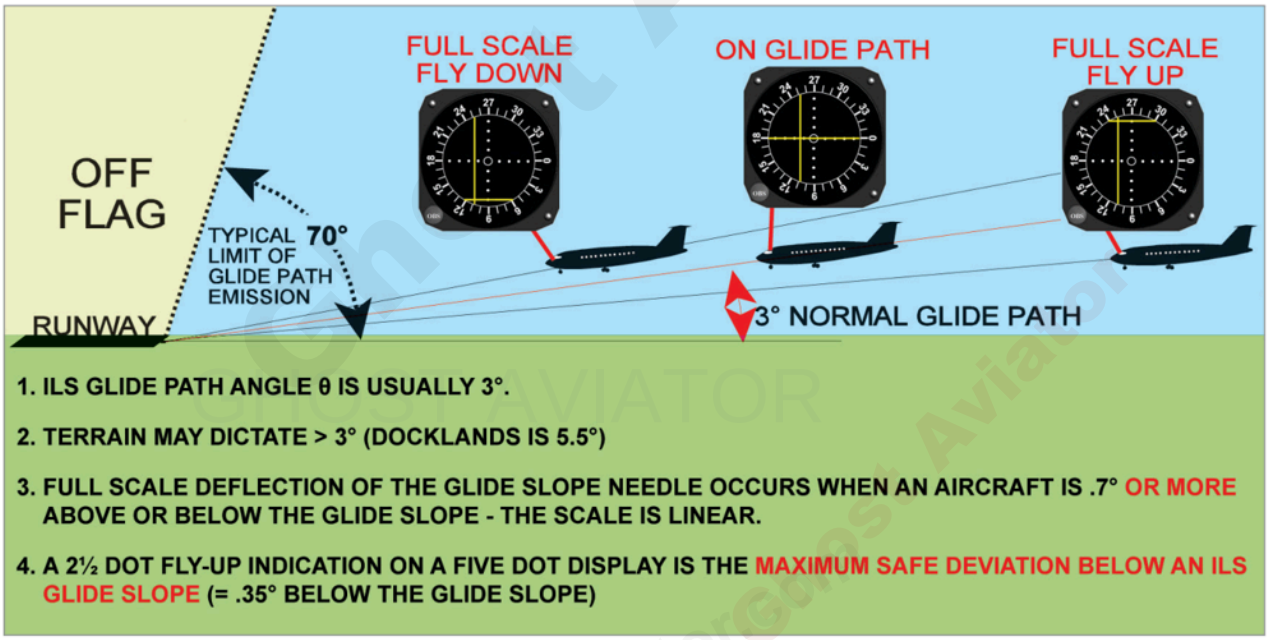


Figure 9.13 Glide path

Fig 9.13: Glide path indications — fly up/fly down

⚡ Reading ILS CDI Summary

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What you see	What it means	Action
Localizer bar right	Aircraft LEFT of centreline	Turn RIGHT
Localizer bar left	Aircraft RIGHT of centreline	Turn LEFT
GP needle high (fly up)	Aircraft BELOW glide slope	Pitch UP / reduce descent
GP needle low (fly down)	Aircraft ABOVE glide slope	Pitch DOWN / increase descent

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9. ILS Categories (ICAO)

📐 Facility Performance Categories (Ground Installation)

Facility Cat	Glide Path Height at Reference Datum
Category I	Guidance to ≤ **200 ft** (60 m) above threshold horizontal plane
Category II	Guidance to ≤ **50 ft** (15 m) above threshold horizontal plane
Category III	Guidance from coverage limit to and along the **runway surface**

📐 Operational Performance Categories (Aircrew/Aircraft)

Operational Cat	Decision Height (DH)	RVR
Cat I	≥ **200 ft** (60 m)	≥ **550 m**
Cat II	≥ **100 ft** (30 m), < 200 ft	≥ **300 m**
Cat IIIA	< 100 ft (30 m) or **no DH**	≥ **200 m**
Cat IIIB	< 50 ft (15 m) or **no DH**	≥ **75 m** but < 200 m

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Cat IIIC	**No DH**	**No RVR limitation**

⚡ Memory Aid — ILS Categories

Cat I = 200ft / 550m | Cat II = 100ft / 300m | IIIA = 0ft / 200m | IIIB = 0ft / 75m | IIIC = Zero Zero

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10. Errors, Accuracy & Protected Areas

⚠ Sources of ILS Error

- **Beam bends**

  : Caused by atmospheric conditions — smooth, slow deflections

- **Scalloping**

  : Caused by reflections → rapid needle fluctuations impossible to follow

- **Beam noise**

  : Generated by transmitter interference

Cross-check with other navaids (DME, altimeter, markers) and monitor ident continuously.

📐 ILS Protected Areas

Area	Definition	When Enforced
Critical Area	Defined area around LLZ and GP antennae where vehicles/aircraft excluded	**ALL ILS operations** (any category)
Sensitive Area	Extends beyond critical area; controls parking/movement of vehicles/aircraft	**Low visibility operations only**
Cat II/III Hold	Pre-take-off holding points further from runway than in good weather	Cat II and III ops; marked with signs and red stop bars

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⚡ Other Factors Affecting Accuracy

- **Weather**

  : Snow and heavy rain attenuates ILS signals, reducing range and accuracy

- **FM Broadcasts**

  : FM stations below 108 MHz can overspill into radio nav band. FM suppression circuits mandatory in ILS receivers since late 1990s.

- **Serviceability check**

  : Ensure failure flags not visible; monitor ident throughout approach

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11. ILS Calculations

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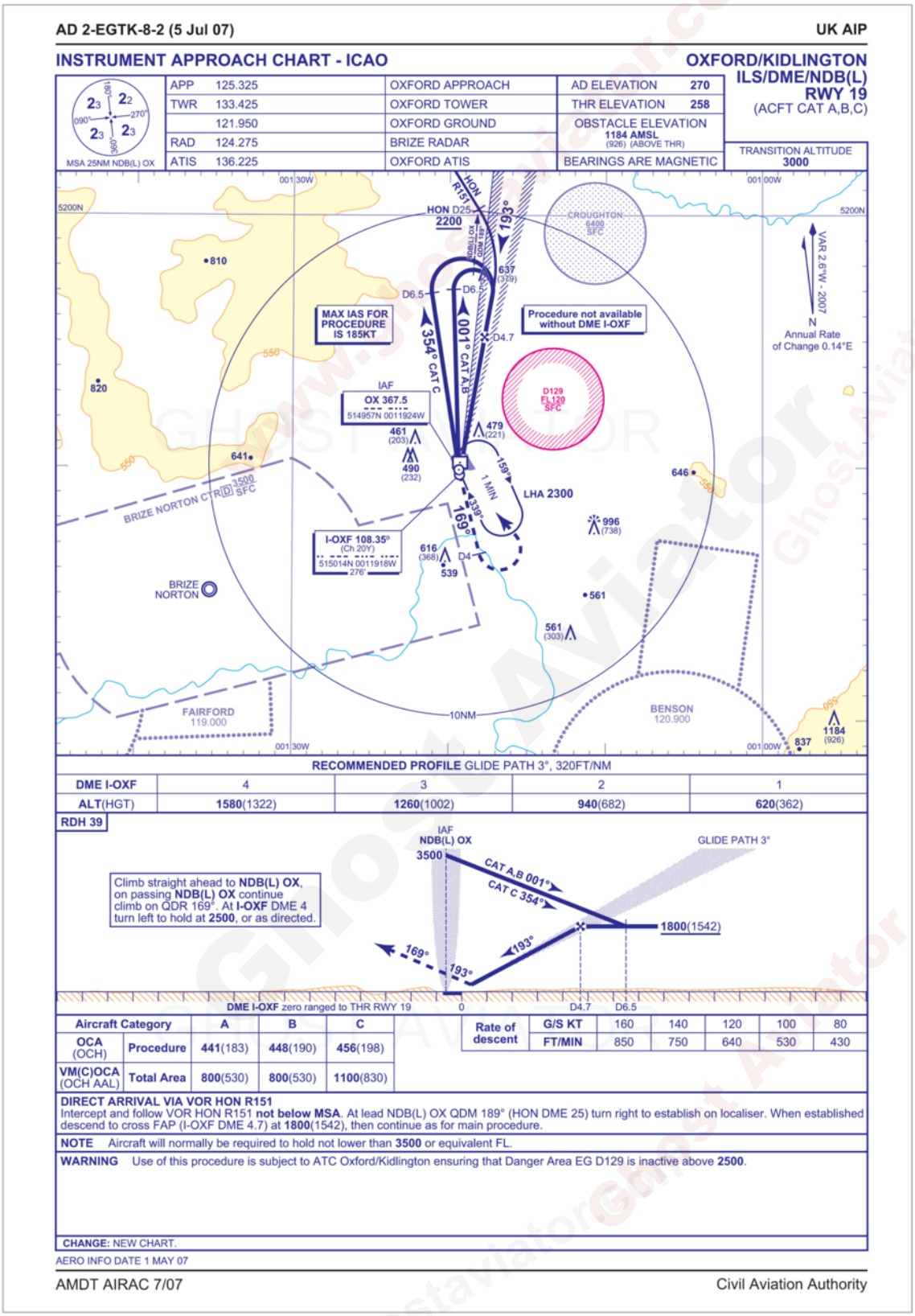


Fig 9.14: ILS approach chart — showing initial, intermediate and final approach segments

Approach Chart Segments

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Segment	Between	Notes
Initial approach	From entry to IAF (Initial Approach Fix)	Sector Safety Altitudes (SSA) apply
Intermediate	IAF to FAF (Final Approach Fix)	Intercept localizer and configure aircraft
Final approach	After FAF to DH/touchdown	ILS guidance; check markers/DME

Height Check on Glide Path (1:60 Rule)

✓ Formula: Height on GP

$$\text{Height (ft)} = \text{GP angle} \times \text{Range (NM)} \times 100$$

(Simplified from the 1:60 rule: GP angle × range/60 × 6076 ft ≈ GP angle × range × 100)

✓ Formula: Rate of Descent (ROD) for 3° GP

$$\text{ROD (fpm)} = 5 \times \text{Groundspeed (kt)}$$

For other GP angles: calculate for 3° then ÷3 × actual angle.

⚡ Worked Example

Aircraft at 4 NM from touchdown, 3° GP, groundspeed 150 kt:

Expected height

= 3 × 4 × 100 = **1200 ft** (exact trig = 1274 ft)

ROD required

= 5 × 150 = **750 fpm** (exact trig = 796 fpm)

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12. ILS Summary

Component	Details
Localizer	VHF 108–111.975 MHz, 40 channels (odd tenths), antenna at upwind end
Glide Path	UHF 329.15–335 MHz, frequency paired, antenna abeam touchdown
Markers	75 MHz, fan-shaped vertical radiation, OM/MM/IM
DME	Frequency paired, zero-referenced to threshold, replaces/supplements markers
Ident	2–3 Morse letters, 7 groups/min, first letter "I"; suppressed when u/s; continuous tone = maintenance
OM	Blue, 2 dashes/sec, 400 Hz, 6.5–11.1 km from threshold
MM	Amber, alt. dots & dashes 3/sec, 1300 Hz, 1050 m from threshold
IM	White, 6 dots/sec, 3000 Hz, 75–450 m from threshold
Monitor triggers	LLZ: >35/25/20 ft shift; GP: >0.075θ change; power >50% reduction
LLZ coverage	25 NM/±10°, 17 NM/±35°
GP coverage	10 NM/±8°, 0.45θ to 1.75θ vertical
LLZ principle	LH lobe 90 Hz (yellow), RH lobe 150 Hz (blue); DDM=0 on %
GP principle	Upper lobe 90 Hz, lower lobe 150 Hz; DDM=0 on GP
False GP	First false at ~2× GP angle (6° for standard 3° GP)
CDI localizer	0.5°/dot, FSD ±2.5°
CDI glide path	0.14°/dot, FSD ±0.7°; max safe dev = 2.5 dots fly up
Height formula	GP angle × range (NM) × 100 = height (ft)
ROD formula (3°)	5 × groundspeed (kt) = ROD (fpm)

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13. Practice Questions

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⚡ Instructions

Click “Show Answer” to reveal the correct answer and explanation.

Q1. The coverage of an ILS localizer extends to __ either side of the on-course line out to a range of ___ NM.

(a) 10°, 35

(b) 35°, 10

(c) 35°, 17

(d) 25°, 25

Answer: (c)

Localizer: **35° from centreline out to 17 NM**; ±10° out to 25 NM. The coverage narrows to 17 NM within the wider ±35° sector.

Q2. The upper and lower limits of an ILS glide path transmitter having a 3.5° glide slope are:

(a) 6.125° – 1.575°

(b) 7.700° – 1.225°

(c) 5.250° – 1.350°

(d) 3.850° – 3.150°

Answer: (a)

Upper = 1.75 × 3.5° = **6.125°**; Lower = 0.45 × 3.5° = **1.575°**.

Q3. The minimum angle at which a false glide path is likely to be encountered on a 3° glide path is:

(a) 6 degrees

(b) 5.35 degrees

(c) normal glide slope times 1.75

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(d) normal glide slope times 0.70

> **Answer: (a)**
> First false GP ≈ **twice the true GP angle** = 2 × 3° = **6°**. False GPs always occur above the true glide slope.

Q4. The visual and aural indications when overflying an ILS middle marker are:

(a) continuous low pitched dashes with blue light

(b) continuous high pitched dots with amber light

(c) alternating medium pitch dots and dashes with amber light

(d) one Morse letter with white light

> **Answer: (c)**
> MM = **Amber** light + **alternating dots and dashes** at 3 per second + **1300 Hz medium pitch** audio.

Q5. An aircraft on ILS approach is receiving stronger 150 Hz signals than 90 Hz. Correct actions to reach centreline and glide path:

(a) DOWN and LEFT

(b) UP and LEFT

(c) UP and RIGHT

(d) DOWN and RIGHT

> **Answer: (b)**
> Stronger 150 Hz on localizer = aircraft in RIGHT (blue) lobe = aircraft is RIGHT of centreline → fly **LEFT**.
> Stronger 150 Hz on GP = aircraft BELOW glide slope (lower lobe = 150 Hz) → fly **UP**.
> Combined: **UP and LEFT**.

Q6. Upper and lower limits of ILS glide path with 3.0° glide slope:

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(a) 0.35° / 0.70°

(b) 3.00° / at least 6°

(c) 5.25° / 1.35°

(d) 10.0° / 35.0°

> **Answer: (c)**
> Upper = 1.75 × 3° = **5.25°**; Lower = 0.45 × 3° = **1.35°**.

Q7. A category II ILS installation in the UK:

(a) provides accurate guidance down to 50 ft above the horizontal plane containing the runway threshold

(b) has a steep glide path, normally 7.5°

(c) provides guidance to and along the runway surface

(d) has a false glide path exactly twice the true GP angle

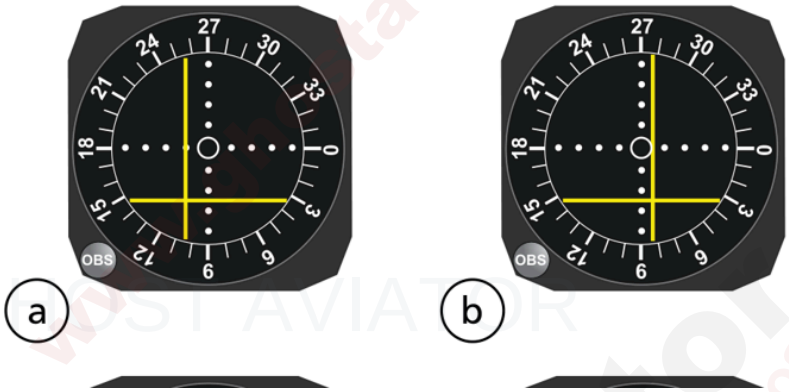
> **Answer: (a)**
> Cat II ILS ground installation provides guidance to the point where the glide path intersects at **50 ft (15 m)** above the threshold horizontal plane.

Q8. Which ILS indicator shows aircraft on final approach left of centreline and at maximum safe deviation below the glide path?

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a. provides accurate guidance down to 50' above the horizontal plane containing the runway threshold
b. has a steep glide path, normally 7.5°
c. provides accurate guidance down to the runway and along the runway after landing
d. has a false glide path that is exactly twice the true glide path angle

8. **Which of these ILS indicators shows an aircraft on final approach left of the centre line and at maximum safe deviation below the glide path?**



(a) (a) — bar right, GP needle centred

(b) (b) — bar left, GP needle high

(c) (c) — bar centred, GP needle centred

(d) (d) — bar right, GP needle high (fly up) at 2.5 dots

Answer: (d)

Aircraft LEFT of centreline → localizer bar deflects **RIGHT**. Below glide path at maximum safe deviation (2.5 dots fly up) → GP needle shows **fly up, 2.5 dots**. This is option (d).

Q9. An aircraft tracking to intercept the ILS localizer inbound but outside the published coverage angle:

(a) will receive false on-course or reverse sense signals

(b) will not normally receive signals

(c) will receive signals without coding

(d) can expect correct indications

Answer: (a)

Outside the coverage sectors, the ILS may give **false on-course or reverse sense**

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signals. The signal exists outside the protected coverage area but cannot be guaranteed to give correct guidance.

Q10. The outer marker of an ILS has visual identification of:

(a) alternating dots and dashes on a blue light

(b) continuous dots at 3 per second, blue light

(c) continuous dashes at 2 per second, amber light

(d) continuous dashes at 2 per second, blue light

Answer: (d)
OM = **Blue** light + **2 dashes per second** (400 Hz, low pitch).

Q11. The specified maximum safe fly-up indication on a 5-dot CDI is:

(a) half full scale deflection above centreline

(b) 2.5 dots fly up

(c) just before full scale deflection

(d) 1.3 dots fly up

Answer: (b)
Maximum safe deviation below glide slope = **2.5 dots fly up** (half the 5-dot full scale). Beyond this, initiate immediate go-around as terrain clearance may be compromised.

Q12. An aircraft using ILS outside coverage sectors — which is correct?

(a) Glide slope false course, localizer intermittent

(b) No transmissions, OFF flags visible

(c) Localizer false course, glide slope reverse sense

(d) Localizer false course/reverse sense; glide slope intermittent/incorrect

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Answer: (d)
Outside coverage: **localizer may give false on-course and reverse sense indications**; **glide slope may give intermittent and incorrect indications**. Answer (d) correctly states this combination.

Q13. Coverage of ILS glide slope in azimuth:

(a) ±8° out to 10 NM

(b) ±10° out to 8 NM

(c) ±12° out to 17 NM

(d) ±35° out to 25 NM

Answer: (a)
GP horizontal coverage: **±8°** azimuth out to at least **10 NM**.

Q14. ILS receivers showing predominant 90 Hz on both localizer and glide slope; runway QDM 264°. Aircraft is:

(a) north of localizer, below glide slope

(b) south of localizer, above glide slope

(c) north of localizer, above glide slope

(d) south of localizer, below glide slope

Answer: (b)
Runway QDM 264° (westward). Standing at the runway facing approach (264°), LEFT is to the south, RIGHT is north.
90 Hz on localizer = aircraft in LEFT (yellow) sector = aircraft is **south** of centreline (left of runway facing west, looking east).
90 Hz on GP = upper lobe = aircraft is **above** glide slope.
Answer: **south, above glide slope**.

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